

INTRODUCTION TO PLANT SCIENCE (HOCHMAN ADLER)



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Intro to Plant Science

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CHAPTER OVERVIEW

1: The Process of Science

Learning Objectives

- Develop a strong understanding of the scientific method and its application in plant science.
- Learn how scientists formulate and test hypotheses.
- Understand the difference between inductive and deductive reasoning.
- Explore how experimentation and peer review contribute to scientific progress.

Science isn't just about memorizing facts—it's a way of exploring and understanding the world around us. In Chapter 1, we will learn that science is a process built on curiosity, creativity, and careful thinking. Scientists ask questions, make observations, and test ideas through experiments. They use both inductive reasoning (looking at patterns to form ideas) and deductive reasoning (using general rules to predict outcomes).

[1.1: The Process of Science](#)

[1.2: Science and Experimentation](#)

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1.1: The Process of Science

Like geology, physics, and chemistry, biology is a science that gathers knowledge about the natural world. Specifically, biology is the study of life. The discoveries of biology are made by a community of researchers who work individually and together using agreed-on methods. In this sense, biology, like all sciences is a social enterprise like politics or the arts.



Figure 1.1.1: Formerly called blue-green algae, the (a) cyanobacteria seen through a light microscope are some of Earth's oldest life forms. These (b) stromatolites along the shores of Lake Thetis in Western Australia are ancient structures formed by the layering of cyanobacteria in shallow waters. (credit a: modification of work by NASA; scale-bar data from Matt Russell; credit b: modification of work by Ruth Ellison)

The methods of science include careful observation, record keeping, logical and mathematical reasoning, experimentation, and submitting conclusions to the scrutiny of others. Science also requires considerable imagination and creativity; a well-designed experiment is commonly described as elegant, or beautiful. Like politics, science has considerable practical implications and some science is dedicated to practical applications, such as the prevention of disease (Figure 1.1.2). Other science proceeds largely motivated by curiosity. Whatever its goal, there is no doubt that science, including biology, has transformed human existence and will continue to do so.

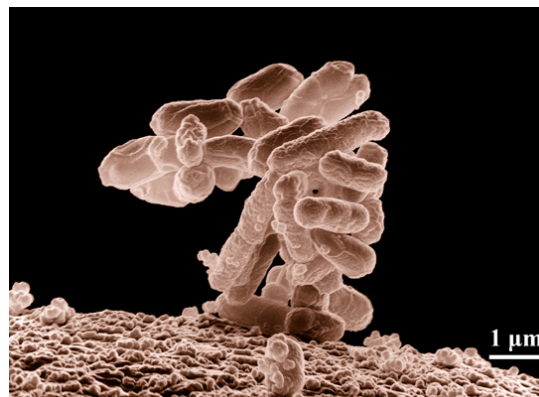


Figure 1.1.2: Biologists may choose to study *Escherichia coli* (*E. coli*), a bacterium that is a normal resident of our digestive tracts but which is also sometimes responsible for disease outbreaks. In this micrograph, the bacterium is visualized using a scanning electron microscope and digital colorization. (credit: Eric Erbe; digital colorization by Christopher Pooley, USDA-ARS)

The Nature of Science

Biology is a science, but what exactly is science? What does the study of biology share with other scientific disciplines? Science (from the Latin *scientia*, meaning "knowledge") can be defined as knowledge about the natural world.

Science is a very specific way of learning, or knowing, about the world. The history of the past 500 years demonstrates that science is a very powerful way of knowing about the world; it is largely responsible for the technological revolutions that have taken place during this time. There are however, areas of knowledge and human experience that the methods of science cannot be applied to. These include such things as answering purely moral questions, aesthetic questions, or what can be generally categorized as spiritual questions. Science has cannot investigate these areas because they are outside the realm of material phenomena, the phenomena of matter and energy, and cannot be observed and measured.

The scientific method is a method of research with defined steps that include experiments and careful observation. The steps of the scientific method will be examined in detail later, but one of the most important aspects of this method is the testing of hypotheses. A hypothesis is a suggested explanation for an event, which can be tested. Hypotheses, or tentative explanations, are generally produced within the context of a scientific theory. A scientific theory is a generally accepted, thoroughly tested and confirmed explanation for a set of observations or phenomena. Scientific theory is the foundation of scientific knowledge. In addition, in many scientific disciplines (less so in biology) there are scientific laws, often expressed in mathematical formulas, which describe how elements of nature will behave under certain specific conditions. There is not an evolution of hypotheses through theories to laws as if they represented some increase in certainty about the world. Hypotheses are the day-to-day material that scientists work with and they are developed within the context of theories. Laws are concise descriptions of parts of the world that are amenable to formulaic or mathematical description.

Natural Sciences

What would you expect to see in a museum of natural sciences? Frogs? Plants? Dinosaur skeletons? Exhibits about how the brain functions? A planetarium? Gems and minerals? Or maybe all of the above? Science includes such diverse fields as astronomy, biology, computer sciences, geology, logic, physics, chemistry, and mathematics (Figure 1.1.3). However, those fields of science related to the physical world and its phenomena and processes are considered natural sciences. Thus, a museum of natural sciences might contain any of the items listed above.

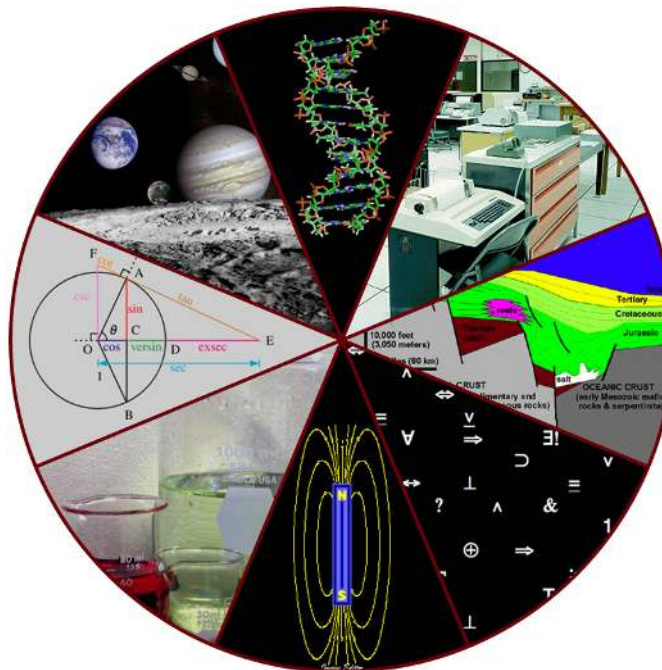


Figure 1.1.3: Some fields of science include astronomy, biology, computer science, geology, logic, physics, chemistry, and mathematics. (credit: "Image Editor"/Flickr)

There is no complete agreement when it comes to defining what the natural sciences include. For some experts, the natural sciences are astronomy, biology, chemistry, earth science, and physics. Other scholars choose to divide natural sciences into life sciences, which study living things and include biology, and physical sciences, which study nonliving matter and include astronomy, physics, and chemistry. Some disciplines such as biophysics and biochemistry build on two sciences and are interdisciplinary.

Scientific Inquiry

One thing is common to all forms of science: an ultimate goal "to know." Curiosity and inquiry are the driving forces for the development of science. Scientists seek to understand the world and the way it operates. Two methods of logical thinking are used: inductive reasoning and deductive reasoning.

Inductive reasoning is a form of logical thinking that uses related observations to arrive at a general conclusion. This type of reasoning is common in descriptive science. A life scientist such as a biologist makes observations and records them. These data can be qualitative (descriptive) or quantitative (consisting of numbers), and the raw data can be supplemented with drawings, pictures, photos, or videos. From many observations, the scientist can infer conclusions (inductions) based on evidence. Inductive

reasoning involves formulating generalizations inferred from careful observation and the analysis of a large amount of data. Brain studies often work this way. Many brains are observed while people are doing a task. The part of the brain that lights up, indicating activity, is then demonstrated to be the part controlling the response to that task.

Deductive reasoning or deduction is the type of logic used in hypothesis-based science. In deductive reasoning, the pattern of thinking moves in the opposite direction as compared to inductive reasoning. Deductive reasoning is a form of logical thinking that uses a general principle or law to forecast specific results. From those general principles, a scientist can extrapolate and predict the specific results that would be valid as long as the general principles are valid. For example, a prediction would be that if the climate is becoming warmer in a region, the distribution of plants and animals should change. Comparisons have been made between distributions in the past and the present, and the many changes that have been found are consistent with a warming climate. Finding the change in distribution is evidence that the climate change conclusion is a valid one.

Both types of logical thinking are related to the two main pathways of scientific study: descriptive science and hypothesis-based science. Descriptive (or discovery) science aims to observe, explore, and discover, while hypothesis-based science begins with a specific question or problem and a potential answer or solution that can be tested. The boundary between these two forms of study is often blurred, because most scientific endeavors combine both approaches. Observations lead to questions, questions lead to forming a hypothesis as a possible answer to those questions, and then the hypothesis is tested. Thus, descriptive science and hypothesis-based science are in continuous dialogue.

Hypothesis Testing

Biologists study the living world by posing questions about it and seeking science-based responses. This approach is common to other sciences as well and is often referred to as the scientific method. The scientific method was used even in ancient times, but it was first documented by England's Sir Francis Bacon (1561–1626) (Figure 1.1.4), who set up inductive methods for scientific inquiry. The scientific method is not exclusively used by biologists but can be applied to almost anything as a logical problem-solving method.



Figure 1.1.4: Sir Francis Bacon is credited with being the first to document the scientific method.

The scientific process typically starts with an observation (often a problem to be solved) that leads to a question. Let's think about a simple problem that starts with an observation and apply the scientific method to solve the problem. One Monday morning, a student arrives at class and quickly discovers that the classroom is too warm. That is an observation that also describes a problem: the classroom is too warm. The student then asks a question: "Why is the classroom so warm?"

Recall that a hypothesis is a suggested explanation that can be tested. To solve a problem, several hypotheses may be proposed. For example, one hypothesis might be, "The classroom is warm because no one turned on the air conditioning." But there could be

other responses to the question, and therefore other hypotheses may be proposed. A second hypothesis might be, “The classroom is warm because there is a power failure, and so the air conditioning doesn’t work.”

Once a hypothesis has been selected, a prediction may be made. A prediction is similar to a hypothesis but it typically has the format “If . . . then” For example, the prediction for the first hypothesis might be, “*If* the student turns on the air conditioning, *then* the classroom will no longer be too warm.”

A hypothesis must be testable to ensure that it is valid. For example, a hypothesis that depends on what a bear thinks is not testable, because it can never be known what a bear thinks. It should also be falsifiable, meaning that it can be disproven by experimental results. An example of an unfalsifiable hypothesis is “Botticelli’s *Birth of Venus* is beautiful.” There is no experiment that might show this statement to be false. To test a hypothesis, a researcher will conduct one or more experiments designed to eliminate one or more of the hypotheses. This is important. A hypothesis can be disproven, or eliminated, but it can never be proven. Science does not deal in proofs like mathematics. If an experiment fails to disprove a hypothesis, then we find support for that explanation, but this is not to say that down the road a better explanation will not be found, or a more carefully designed experiment will be found to falsify the hypothesis.

Each experiment will have one or more variables and one or more controls. A variable is any part of the experiment that can vary or change during the experiment. A control is a part of the experiment that does not change. Look for the variables and controls in the example that follows. As a simple example, an experiment might be conducted to test the hypothesis that phosphate limits the growth of algae in freshwater ponds. A series of artificial ponds are filled with water and half of them are treated by adding phosphate each week, while the other half are treated by adding a salt that is known not to be used by algae. The variable here is the phosphate (or lack of phosphate), the experimental or treatment cases are the ponds with added phosphate and the control ponds are those with something inert added, such as the salt. Just adding something is also a control against the possibility that adding extra matter to the pond has an effect. If the treated ponds show lesser growth of algae, then we have found support for our hypothesis. If they do not, then we reject our hypothesis. Be aware that rejecting one hypothesis does not determine whether or not the other hypotheses can be accepted; it simply eliminates one hypothesis that is not valid (Figure 1.1.5). Using the scientific method, the hypotheses that are inconsistent with experimental data are rejected.

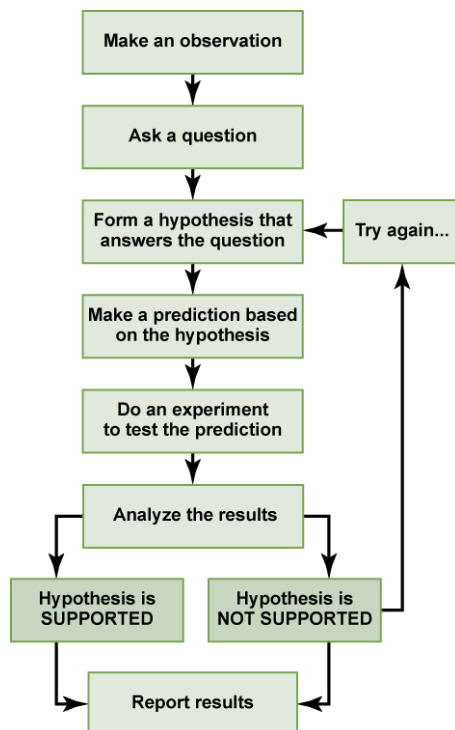


Figure 1.1.5: The scientific method is a series of defined steps that include experiments and careful observation. If a hypothesis is not supported by data, a new hypothesis can be proposed.

Example 1.1.1

In the example below, the scientific method is used to solve an everyday problem. Which part in the example below is the hypothesis? Which is the prediction? Based on the results of the experiment, is the hypothesis supported? If it is not supported, propose some alternative hypotheses.

1. My toaster doesn't toast my bread.
2. Why doesn't my toaster work?
3. There is something wrong with the electrical outlet.
4. If something is wrong with the outlet, my coffeemaker also won't work when plugged into it.
5. I plug my coffeemaker into the outlet.
6. My coffeemaker works.

Solution

The hypothesis is #3 (there is something wrong with the electrical outlet), and the prediction is #4 (if something is wrong with the outlet, then the coffeemaker also won't work when plugged into the outlet). The original hypothesis is not supported, as the coffee maker works when plugged into the outlet. Alternative hypotheses may include (1) the toaster might be broken or (2) the toaster wasn't turned on.

In practice, the scientific method is not as rigid and structured as it might at first appear. Sometimes an experiment leads to conclusions that favor a change in approach; often, an experiment brings entirely new scientific questions to the puzzle. Many times, science does not operate in a linear fashion; instead, scientists continually draw inferences and make generalizations, finding patterns as their research proceeds. Scientific reasoning is more complex than the scientific method alone suggests.

Basic and Applied Science

The scientific community has been debating for the last few decades about the value of different types of science. Is it valuable to pursue science for the sake of simply gaining knowledge, or does scientific knowledge only have worth if we can apply it to solving a specific problem or bettering our lives? This question focuses on the differences between two types of science: basic science and applied science.

Basic science or “pure” science seeks to expand knowledge regardless of the short-term application of that knowledge. It is not focused on developing a product or a service of immediate public or commercial value. The immediate goal of basic science is knowledge for knowledge's sake, though this does not mean that in the end it may not result in an application.

In contrast, applied science or “technology,” aims to use science to solve real-world problems, making it possible, for example, to improve a crop yield, find a cure for a particular disease, or save animals threatened by a natural disaster. In applied science, the problem is usually defined for the researcher.

Some individuals may perceive applied science as “useful” and basic science as “useless.” A question these people might pose to a scientist advocating knowledge acquisition would be, “What for?” A careful look at the history of science, however, reveals that basic knowledge has resulted in many remarkable applications of great value. Many scientists think that a basic understanding of science is necessary before an application is developed; therefore, applied science relies on the results generated through basic science. Other scientists think that it is time to move on from basic science and instead to find solutions to actual problems. Both approaches are valid. It is true that there are problems that demand immediate attention; however, few solutions would be found without the help of the knowledge generated through basic science.

One example of how basic and applied science can work together to solve practical problems occurred after the discovery of DNA structure led to an understanding of the molecular mechanisms governing DNA replication. Strands of DNA, unique in every human, are found in our cells, where they provide the instructions necessary for life. During DNA replication, new copies of DNA are made, shortly before a cell divides to form new cells. Understanding the mechanisms of DNA replication enabled scientists to develop laboratory techniques that are now used to identify genetic diseases, pinpoint individuals who were at a crime scene, and determine paternity. Without basic science, it is unlikely that applied science would exist.

Another example of the link between basic and applied research is the Human Genome Project, a study in which each human chromosome was analyzed and mapped to determine the precise sequence of DNA subunits and the exact location of each gene. (The gene is the basic unit of heredity; an individual's complete collection of genes is his or her genome.) Other organisms have also been studied as part of this project to gain a better understanding of human chromosomes. The Human Genome Project (Figure

1.1.6) relied on basic research carried out with non-human organisms and, later, with the human genome. An important end goal eventually became using the data for applied research seeking cures for genetically related diseases.



Figure 1.1.6: The Human Genome Project was a 13-year collaborative effort among researchers working in several different fields of science. The project was completed in 2003. (credit: the U.S. Department of Energy Genome Programs)

While research efforts in both basic science and applied science are usually carefully planned, it is important to note that some discoveries are made by serendipity, that is, by means of a fortunate accident or a lucky surprise. Penicillin was discovered when biologist Alexander Fleming accidentally left a petri dish of *Staphylococcus* bacteria open. An unwanted mold grew, killing the bacteria. The mold turned out to be *Penicillium*, and a new antibiotic was discovered. Even in the highly organized world of science, luck—when combined with an observant, curious mind—can lead to unexpected breakthroughs.

Reporting Scientific Work

Whether scientific research is basic science or applied science, scientists must share their findings for other researchers to expand and build upon their discoveries. Communication and collaboration within and between sub disciplines of science are key to the advancement of knowledge in science. For this reason, an important aspect of a scientist's work is disseminating results and communicating with peers. Scientists can share results by presenting them at a scientific meeting or conference, but this approach can reach only the limited few who are present. Instead, most scientists present their results in peer-reviewed articles that are published in scientific journals. Peer-reviewed articles are scientific papers that are reviewed, usually anonymously by a scientist's colleagues, or peers. These colleagues are qualified individuals, often experts in the same research area, who judge whether or not the scientist's work is suitable for publication. The process of peer review helps to ensure that the research described in a scientific paper or grant proposal is original, significant, logical, and thorough. Grant proposals, which are requests for research funding, are also subject to peer review. Scientists publish their work so other scientists can reproduce their experiments under similar or different conditions to expand on the findings. The experimental results must be consistent with the findings of other scientists.

There are many journals and the popular press that do not use a peer-review system. A large number of online open-access journals, journals with articles available without cost, are now available many of which use rigorous peer-review systems, but some of which do not. Results of any studies published in these forums without peer review are not reliable and should not form the basis for other scientific work. In one exception, journals may allow a researcher to cite a personal communication from another researcher about unpublished results with the cited author's permission.

Summary

Biology is the science that studies living organisms and their interactions with one another and their environments. Science attempts to describe and understand the nature of the universe in whole or in part. Science has many fields; those fields related to the physical world and its phenomena are considered natural sciences.

A hypothesis is a tentative explanation for an observation. A scientific theory is a well-tested and consistently verified explanation for a set of observations or phenomena. A scientific law is a description, often in the form of a mathematical formula, of the behavior of an aspect of nature under certain circumstances. Two types of logical reasoning are used in science. Inductive reasoning uses results to produce general scientific principles. Deductive reasoning is a form of logical thinking that predicts results by applying general principles. The common thread throughout scientific research is the use of the scientific method. Scientists present their results in peer-reviewed scientific papers published in scientific journals.

Science can be basic or applied. The main goal of basic science is to expand knowledge without any expectation of short-term practical application of that knowledge. The primary goal of applied research, however, is to solve practical problems.

Glossary

applied science

a form of science that solves real-world problems

basic science

science that seeks to expand knowledge regardless of the short-term application of that knowledge

control

a part of an experiment that does not change during the experiment

deductive reasoning

a form of logical thinking that uses a general statement to forecast specific results

descriptive science

a form of science that aims to observe, explore, and find things out

falsifiable

able to be disproven by experimental results

hypothesis

a suggested explanation for an event, which can be tested

hypothesis-based science

a form of science that begins with a specific explanation that is then tested

inductive reasoning

a form of logical thinking that uses related observations to arrive at a general conclusion

life science

a field of science, such as biology, that studies living things

natural science

a field of science that studies the physical world, its phenomena, and processes

peer-reviewed article

a scientific report that is reviewed by a scientist's colleagues before publication

physical science

a field of science, such as astronomy, physics, and chemistry, that studies nonliving matter

science

knowledge that covers general truths or the operation of general laws, especially when acquired and tested by the scientific method

scientific law

a description, often in the form of a mathematical formula, for the behavior of some aspect of nature under certain specific conditions

scientific method

a method of research with defined steps that include experiments and careful observation

scientific theory

a thoroughly tested and confirmed explanation for observations or phenomena

variable

a part of an experiment that can vary or change

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1.2: Science and Experimentation

Learning objectives

- Describe why science is considered a discipline of philosophy.
- Summarize the four basic types of experiments.
- Apply the principles of experimental design in this course and in your daily life.

Thinking about science

The primary goal of this section is to help you think about the nature of science. You might be taking this course to fulfill an undergraduate requirement for a biology course. This course fulfills that requirement because we investigate the process behind using science as a way of learning about the natural world around us. If you're starting down the path to becoming a plant scientist, understanding the nature of science will be essential for you in your career.

Regardless of whether you're going to pursue a career as a scientist, now is a good time to reflect on the nature of science, and to understand how scientific thinking can become a strategy for resolving many issues that you confront during daily life.

Watch this video about connecting science and experimentation to real life:

Scientific inquiry

While “science” is a word commonly used in our culture, in popular use it is rarely spoken of as a philosophy. By identifying science as a philosophy we are taking an [epistemic view](#), one focusing on how knowledge is acquired.

At its core, science is a mode of inquiry: a way of acquiring new knowledge about the world around us and a strategy for understanding the inner workings of elements in that world. Scientists believe that if we follow the principles of this philosophy we will continue to expand our knowledge about how things work in the world around us. This systematic approach is called the “scientific method.”

There are two key steps in the scientific method:

- Hypothesis building through reflective observation.
- Hypothesis testing through experimentation.

A “hypothesis” is a question or proposed explanation made on the basis of limited evidence and used as a starting point for experimentation. Experimentation is commonly equated with science—rightly so, because hypotheses are evaluated on the basis of evidence generated through experiments. Experimentation, however, isn't the whole story. Science—including the development and testing of new hypotheses—is also a creative endeavor.

Watch this video about scientific inquiry:

Scientific inquiry has generated a vast body of knowledge about the world around us. Your school science classes might have required you to memorize facts and relationships, and pay attention to detail. Sometimes such memorization leads students to believe that science is just an accumulation of facts rather than the process behind discovering all of that information.

Scientific discovery builds on what is already known. Even the most accomplished scientists initially approach a problem by learning what is already known. Armed with that information, they then apply their own creativity to form new hypotheses about something they have observed, and design experiments to test those hypotheses. They also communicate their results publicly so that others can benefit from their work and have the opportunity to challenge conclusions. In this way, science builds on itself.

The foundational knowledge you learn in science classes prepares you to develop and test hypotheses and to make new discoveries of your own. While a good memory may help you pass a science class, you will absorb a body of knowledge more effectively when you learn how facts fit and work together in systems rather than learning through the brute force of memorization.

In this section we work from the point of view that science is a way of acquiring knowledge—a mode of inquiry—and that this mode of inquiry follows a process called the scientific method. Those who follow the philosophy of science:

- Use it to understand how the natural world works.
- Start by learning what is already known.

- Carefully observe the subjects of their scientific inquiry and look for details about form, function, and interaction with the environment.
- Develop hypotheses about the inner workings of natural phenomena not yet understood.
- Test their hypotheses by making observations, conducting experiments and collecting and evaluating evidence.
- Communicate with others about their hypotheses, experiments, and the outcomes of their studies so that others can repeat, validate, and build upon their work.

Although science is typically used to understand how the natural world works, it is also regularly applied to the development of new technologies that are based on these natural phenomena and to the solving of problems associated with the natural world.

Putting the scientific method to work

As noted, the scientific method relies on building hypotheses and then testing them through experimentation. In the lab section of this course you will develop hypotheses about the effects of various treatments on propagation success and then conduct experiments to test those hypotheses. Because experimentation is such a key component of the scientific method, we'll spend time characterizing and examining four types of experimentation and explore whether they are part of the scientific method. While each is valuable when applied in the right circumstances, only one clearly follows each step of the scientific method to uncover new knowledge about the natural world.

Types of experiments

The types of experimentation we will cover are:

- Demonstration
- Evaluation
- Exploration
- Discovery

Demonstration experiments



Many experiments conducted in lab courses are demonstration experiments. Photo by [Salish Sea Expeditions](#). CC BY-NC-ND 2.0

Demonstration experiments are a classic method used in educational settings to help students learn and understand known relationships already discovered by others. Learners will usually have had prior exposure to the relationships through preliminary observations, lectures, reading, and discussions, and will have some sense of what the experimental outcome might be.

Good demonstration experiments actively involve the learner, who manipulates the experimental materials, applies the treatments, and observes the outcomes, then gathers, analyzes, and interprets the resulting data. Poor demonstration experiments, in contrast, make learners only passive witnesses to something done by an expert at the front of the classroom.

In the plant propagation labs for this course, you will be actively engaged in demonstration experiments. Although you won't be creating new knowledge, the knowledge will likely be new to you. The hands-on experience of conducting the experiments will help you to learn the concepts more effectively than if you only read a textbook or listened to a lecture. The techniques you learn and use in demonstration experiments often contribute to the learning experience as much as the relationships revealed at the

experiment's conclusion. Employing these techniques will help you gain an understanding of many biological functions, such as the production of adventitious roots and mechanisms for seed dispersal.

While demonstration experiments are valuable for actively learning a body of scientific knowledge previously discovered and communicated by others, the experience is specifically orchestrated for teaching and learning, not for the discovery of new information. Yet since the knowledge is new to the learner, it can still bring the joy of personal discovery and a sense of accomplishment.

In summary, demonstration experiments:

- Are designed for teaching and learning.
- Address relationships that may be new to you, but are otherwise known.
- In their best forms, actively involve the learner.
- May emphasize experimental techniques, in addition to outcomes, as part of the learning experience.
- Are not the types of experiments that are at the core of practicing science as a way to uncover new knowledge.

Evaluation experiments

Evaluation experiments are designed to help us make decisions, and to choose from a number of options. They might, for instance, help us determine the efficacy of a new treatment relative to a known treatment, or decide on further experimentation. An evaluation experiment will highlight a compound, a technique, a piece of equipment, or an organism, and will include a control and/or other alternatives.

Evaluation experiments are common in horticultural and agronomic research, where the purpose of the experiment is to identify, for example, the best cultivar, production method, pest control, fertility regime, or light intensity for growing a crop. Correct experimental design is crucial for assuring that conclusions from the experiment are meaningful and credible.



This field experiment is testing different living mulches between rows of strawberries. Photo by University of Minnesota West Central Research and Outreach Center.

These experiments are typically used in the development of new technologies to identify the best method for the desired purpose (e.g., which pesticides are effective against the target insect, but not harmful to non-target insects). They are not used to discover new knowledge about how the world works, as they typically don't advance our understanding of the natural world. The

information from an evaluation experiment might, however, point the way to additional experimentation that does help us discover new knowledge. This is particularly true if the outcome of an evaluation experiment is unexpected or novel.

In summary, evaluation experiments:

- Are used to help in decision-making.
- Help users choose a winner or determine efficacy relative to other alternatives.
- Are commonly used when evaluating and recommending horticultural production methods.
- Can be useful in solving problems and developing technologies.
- Require proper experimental design (e.g., comparison to a control) for credibility and meaningfulness.

Exploration experiments

Some scientists specialize in observing and cataloging nature, and some, like members of the group that discovered a [new species of hominid, *Homo naledi* in a South African cave](#), aggressively search for previously unknown phenomena. In the botanical realm, such scientists study the diversity of organisms within habitats, discover new species, or are in other ways very skilled in “seeing” nature. Explorer-scientists recognize and appreciate detail and can identify the enormous diversity among plants by comparing characteristics that might be overlooked by others. They may also have the capacity to recognize possible interrelationships among organisms and with habitats, making their work particularly important to science. They might notice, for instance, that a particular species of plant is commonly found in wet areas but not in dry, or that a particular vegetable tastes sweeter when grown at higher altitudes than when grown closer to sea level. They don’t confirm the cause of these relationships, but are the first to notice them.



This scientist is collecting plants in Ecuador to identify unknown species and to determine relatedness to other plants. Photo by Dr. Eric Tepe, University of Cincinnati.

Explorers’ observations are essential to stimulating the development of sound, testable hypotheses. The possible relationships they propose must be tested to determine whether those relationships actually exist, or are artifacts of other effects. Explorers help develop hypotheses, but the work of exploration, cataloging, and seeing possible relationships don’t prove or disprove the hypotheses or necessarily generate new knowledge about relationships. The work does, however, result in new information about the existence of the object or phenomenon itself. An exception is exploration done to test a hypothesis, such as a mission to test the hypothesis that a particular type of ecosystem is required for reproduction of a particular plant species.

Scientists must resist jumping to conclusions based on exploration and observation alone. If you see two people together many times, for example, you might conclude that they are a romantic couple, when in fact they are brother and sister. Relationships hypothesized as a result of exploration and observation must be experimentally tested before they are accepted or rejected.

Exploration experiments uncover new things, many of which can be exciting and eventually change our view of the world. While one of their greatest values is that they lead to the development of new and stronger hypotheses about how the world works, they go so far as to test those hypotheses or fully engage in the cycle of knowledge generation associated with the scientific method. Additional experiments based on this new information are required to put this new information in context and to advance our understanding of how the natural world works.

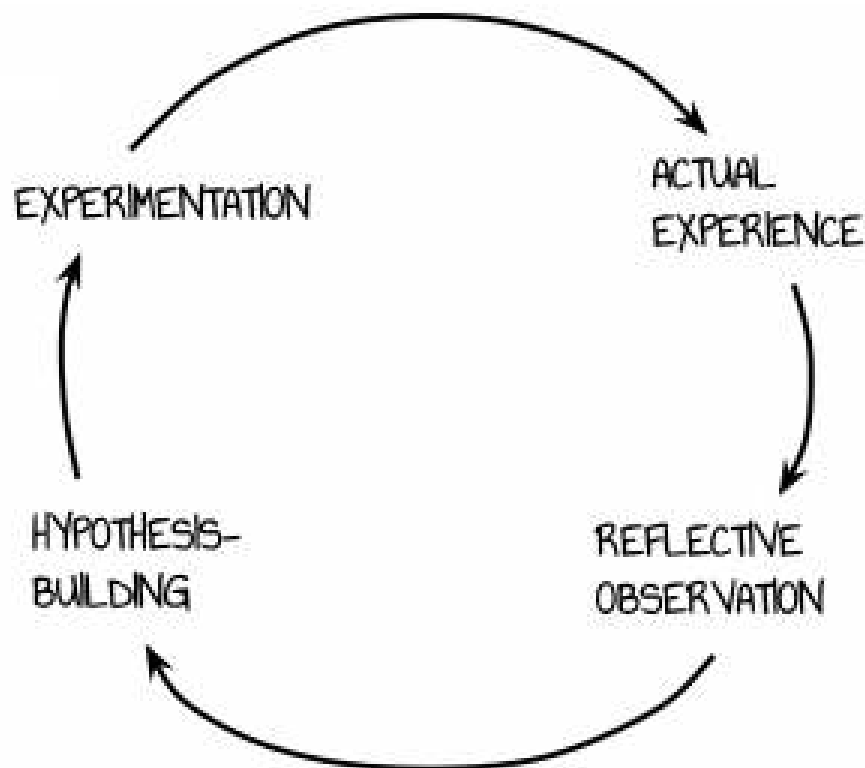
In summary then, exploration experiments:

- Focus on detailed observation of organisms and habitats.
- Increase our knowledge of the natural world.
- Identify potential relationships that need to be tested.
- Are essential to sound and testable hypothesis-building.

Discovery experiments

Discovery experiments are central to the use of the scientific method in tasks ranging from problem solving to the discovery of new knowledge. They focus on uncovering new relationships and solving problems, follow the scientific method, test hypotheses and their predicted outcomes, and utilize a careful design in order to maintain meaningfulness and credibility.

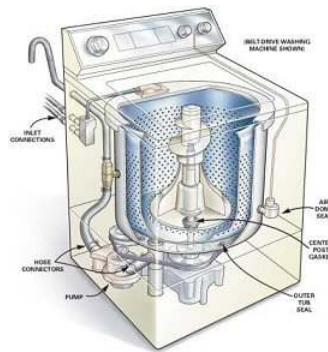
The similarity between the scientific method and [Kolb's Experiential Learning Cycle](#) is not an accident. The scientific method is a practical strategy based on how we sense and experience the world around us and used to solve problems encountered during those experiences.



The scientific method is a great example of the experiential learning theory. Illustration by Emily Tepe.

The diagram above illustrates a combination of the scientific method and Kolb's four-step experiential learning, describing a cyclic process for solving problems that can be applied to disciplines as diverse as molecular biology, global warming, and even appliance repair. While you might initially think that appliance repair doesn't belong in that list, the difference is one of application, not method. Though far removed from the esoteric scientific discoveries we associate with scientific method, appliance repair follows

the same steps. Appliances are often, and quite literally, boxes, where you don't know what is going on inside. But what's going on inside is knowable, and through that knowledge comes repair.



We can apply the scientific method to everyday problems like figuring out why a washing machine isn't working. © HomeTips.com.

The learning/problem solving/scientific process could theoretically start anywhere in Kolb's cycle. But it will likely start with a problem that needs to be solved, something you don't understand but would like to know more about. You become aware that there is a problem or that you lack understanding because you have an experience where you observe something and then step back and said, "I wonder how that works," or perhaps, "why is that broken?" Through observation you develop a sufficiently adequate description of the problem to start doing some research on what is already known.

With a good description of the problem in hand, you can begin to review what is known through the work of others, and think about what might be going on in your situation and how your new understanding can be applied to the problem. This is "reflective observation." It isn't just sitting back and thinking in a vacuum. You need raw material for your mind to work on, and that only comes through the tough task of gathering and engaging with the background information. There is a very important quiet phase in this process when you let your mind assemble and sort through ideas until alternatives begin to emerge that might lead to a solution. Talking with others and sharing ideas is an important part of this quiet phase.

Sometimes the alternatives are no-brainers (blown fuse?), and sometimes they're more creative (residue from the wrong detergent gunking up the water level sensor?). Regardless of their simplicity or complexity, these become hypotheses that need to be tested. The hypothesis-building stage includes both a statement of how something works or why it isn't working, and predictions about what might happen if the hypothesis is true. In appliance repair, for example, the prediction will likely be that the appliance will function normally. In horticultural molecular biology, it might be that you will see accumulation of a particular type of fatty acid in the cotyledons.

You put the hypothesis to the test by designing an experiment that assesses whether your predictions were right. If the outcome doesn't match your prediction, you reject the hypothesis (the fuse was ok, so that wasn't the problem). If the outcome does match your prediction, you tentatively accept the hypothesis pending further observation (when the fuse was replaced the washing machine worked again, so it might have been a blown fuse, but on the other hand maybe it was just because the motor had time to cool down). As with evaluation experimentation, experimental design is important in assuring that the conclusions from the experiment are meaningful and credible.

Experimentation leads to new experiences and an incremental increase in knowledge, and then the cycle begins again.

In summary, discovery experiments:

- Focus on uncovering new relationships and solving problems.
- Follow scientific method.
- Test hypotheses and their predicted outcomes.
- Utilize a careful design in order to maintain meaningfulness and credibility.

Summary

Of the four types of experiments, only the discovery experiments are core to the process of science in the narrow sense of being a way of acquiring new knowledge. The other three types of experimentation are still important; demonstration and evaluation experiments are valuable for learning and decision-making and for technology development, and exploration experiments are essential for developing testable hypotheses. But discovery experiments are core to science.

Remember: the methodology of effective washing machine repair, when applied to what is unknown about the physical world, is the methodology of science. It's not esoteric; it's good appliance repair.

You might argue that, when applied to a broken washing machine, a discovery experiment results in knowledge that is probably already known by those skilled in appliance repair, so it isn't really new knowledge about how the world works. That's a fair criticism. Use of the scientific method can result in new knowledge about how the world works, but whether it uncovers new knowledge depends on the object of experimentation.

? Review questions

An interactive H5P element has been excluded from this version of the text. You can view it online here:
<https://open.lib.umn.edu/horticulture/?p=94#h5p-3>

Experimental design

The methods for designing experiments are carefully studied and often discipline-specific. Methods used in molecular biology, for instance, will be somewhat different from those used in chemistry or in field evaluations of horticultural plants. There are, however, some generalizations we can make about good experimental designs.

Emphasize comparisons

Experiments include more than just one treatment. “Treatment” refers to the factor that you are varying in your experiment—for example, different cultivars of tomato, different fertilizers, or different amounts of light. Experimental designs incorporate comparison of treatments. You usually compare the treatments to one another and often to a control, which is either the application of no treatment or the application of a customary or standard level of treatment.

If you grow a particular type of tomato in your garden, and find that it produces tasty fruit, would you declare it to be the best tomato variety you could grow? Certainly not. You couldn't even say with certainty that it was the best tomato variety you have ever grown (unless it is the only one you have grown). Next year, however, you could grow that tomato as your control, and grow two other varieties that your neighbors like, and compare fruit quality (appearance, flavor, yield, sugar content). You could then say something definitive about the three tomato varieties because you have compared them to each other after growing them next to each other in the same year and environment.

Replicate treatments

The same treatment is applied to more than one “experimental unit”—the object that receives the treatment. In the example above, the tomato plant is the experimental unit, and you would perhaps plant two or three seedlings of each tomato variety rather than just one. Think of a treatment as something like a fertilizer spread on a patch of land. The patch of land is the experimental unit, while the fertilizer is the treatment.

By applying the treatment to more than one experimental unit you can estimate the variation you get when two experimental units are treated the same, and compare this to the variation when experimental units are given different treatments. If the treatments actually differ in their effectiveness, you would expect the variation between experimental units given different treatments to be much greater than the variation between those given the same treatment. This is one of the fundamental ways in which experiments are statistically analyzed and treatments declared significantly different or not.

Randomize treatments

Once you know how many treatments you are going to apply, and how many replications you want, the product of these two quantities ($\# \text{ treatments} \times \# \text{ replications}$) equals the number of experimental units you need. For instance, if you have three fertilizers you want to test, plus a control, you have four treatments. If you want three replications of each treatment, then you $4 \text{ treatments} \times 3 \text{ replications} = 12$ experimental units or patches of land where you will apply the fertilizers. The treatments will be randomly assigned to each experimental unit (patch of land). This is done using a random number table and is not just haphazard picking. Randomization helps minimize any bias you haven't recognized in advance and controlled for in other ways.

? Review questions

1. What are two types of control treatments?
2. Does increasing the number of replications increase the number of treatments or the number of experimental units?
3. Can you think of an example of how randomization can protect against bias?

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